SUPPLEMENTARY MATERIALS FOR "QUERY-FOCUSED EVENT SUMMARIZATION: A DATASET AND BENCHMARK"

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1. Related Work

In this section, we first review related work on QFS datasets, followed by existing studies on QFS methods.

1.1. QFS Datasets.

Early widely used datasets include DUC [2] and TAC [3], both released by NIST to support summarization research. These datasets contain multiple news articles with queries spanning diverse domains such as politics, biography, and disaster events. Although they are targeted at query-focused summarization within thematic corpora, access to these datasets currently requires formal application, making them less convenient for research. Moreover, their limited number of documents and queries prevents them from forming large-scale resources; summaries constructed from only a few dozen documents struggle to comprehensively capture the narrative of thematic events. A dataset with similar scale constraints is TD-QFS [1], a medical dataset covering four topics (asthma, lung cancer, obesity, and Alzheimer's disease), where the queries focus on factors such as causes, treatments, and adolescent impact.

In recent years, research interest in QFS datasets has increased, motivated by users' desire for personalized and query-driven information access. However, few newly proposed QFS datasets are explicitly designed for event summarization. For example, AQuaMuSe [11] simulates search engines generating high-quality paragraphs for user queries based on NQ [12] and Common Crawl, resulting in 5,519 query-focused summaries with an average of 6 documents each. QMSum [27] focuses on query-based multi-domain meeting summarization, containing 1,808 summaries from 232 meetings across academic, product, and committee domains. SQuALITY [25], built on QuALITY [18], is a question-driven dataset of public-domain short stories, including 100 stories with 5 document–query pairs per story, each 3k–6k words in length. WebCiteS [4] is a Chinese dataset designed for attributed query-focused summarization (AQFS), containing 7k human-annotated summaries with citations.However, these datasets are not explicitly designed around events, lack document–query relevance labels, or contain too few documents to ensure comprehensive event coverage—thereby limiting their suitability for real-world, QFES research.

1.2. QFS Methods.

Query-focused summarization (QFS) aims to generate summaries that stay highly relevant to a given query conditioned on a set of input documents. Early work in this area predominantly adopted unsupervised extractive approaches. For example, Wan et al. [24] introduced a manifold-ranking–based sentence extraction method. With the advent of Transformer architectures [22], Transformer-based summarizers have achieved substantial performance gains [19]. In this stage of development, various techniques were further introduced to enhance summary quality. SEGENC [23] enables the decoder to attend jointly to multiple encoded segments using segment encoders. Socratic [17] augments SEGENC with question-driven pretraining to strengthen query grounding. Ranking-based learning has also become widely adopted: Ranker-Generator [15] leverages pairwise and global ranking to select the top-k most relevant content before generation, while Sotudeh et al. [20] jointly train a

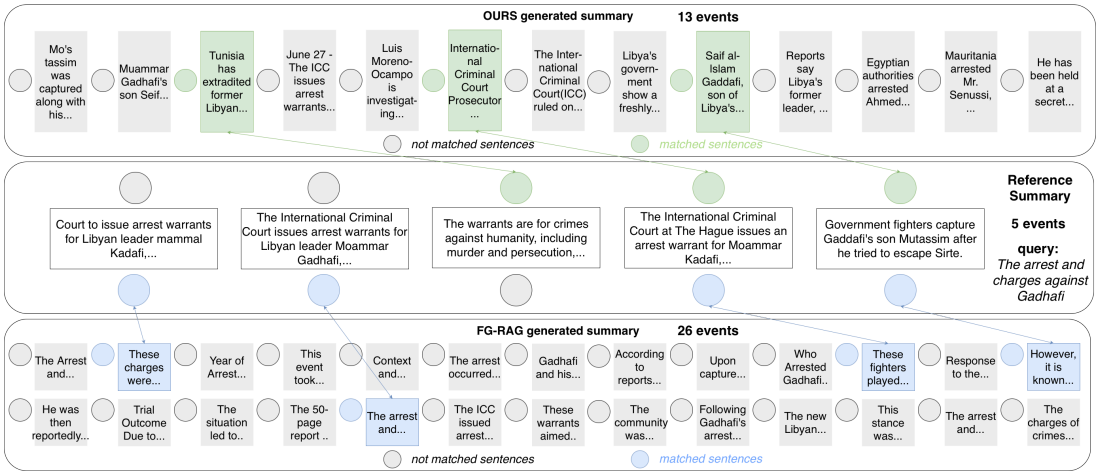


FIGURE 1. Case study of LLM-based evaluation under the query "`The arrest and charges against Gadhafi`".

decoder for both ranking and summarization to improve paragraph prioritization. Nath et al. [16] further introduced reinforcement learning (RL) into QFS, training multiple policy gradient agents based on diverse reward signals.

Beyond the above techniques, graph-based approaches have gained growing attention due to their ability to capture rich structural relations across documents. GraphRAG [5] first constructs a complete knowledge graph from text, performs community detection, then generates hierarchical summaries in a bottom-up fashion. LightRAG [7] combines graph structure with vector representation in a dual-retrieval framework to improve semantic coverage. NaiveRAG [6] segments documents into chunks and retrieves the most relevant ones using dense embeddings. SubGraphRAG [13] enhances retrieval precision via entity-based subgraph expansion, particularly benefiting multi-hop QA settings. FG-RAG [8] further introduces context-aware entity expansion to increase entity coverage within the graph and provide richer supporting context.However, dense semantic retrieval methods are limited by PLM input windows, while graph-based generative models incorporate large amounts of query-irrelevant content, causing noise and overly generic summaries as corpus size grows—revealing a clear gap in scaling QFS to long-document and real-world thematic event settings.

2. CASE STUDY

As shown in Figure 1, this section presents a case study to illustrate large language model–based evaluation examples of our method and FG-RAG when performing on the QFESum dataset. Under a query "`The arrest and charges against Gadhafi`" in the Libya thematic corpus, the reference summary contains a total of 5 sentences. Each sentence corresponds to one event, resulting in a total of five events. Our method generates 13 events. The LLM iterates over the 13 generated events and, for each event, compares it against the 5 events in the reference summary to determine semantic matches, ensuring that each reference event is matched at most once. Among our generated events, 3 events are judged by the LLM to have semantic correspondences with the reference summary, as highlighted with green circles. Accordingly, LLM-Pre = 23.08%, LLM-Rec = 60.00%, and LLM-F1 = 33.34%.

Method + Input Documents	Avg. #Evt.
Reference Summary	21.53
SHC + $\mathcal{D}_i^{\mathrm{r}}$	28.01
TopicTLS + $\mathcal{D}_i^{\mathrm{r}}$	40.48
GraphRAG + $\mathcal{D}_i^{\mathrm{r}}$	16.28
FG-RAG + $\mathcal{D}_i^{\mathrm{r}}$	47.01
UnstructBase + $\mathcal{D}_i^{\mathrm{r}}$	17.69

TABLE 1. Average number of summary events generated by different methods.

In contrast, the FGRAG generates a longer summary, with 26 events in total, of which 4 events are matched as highlighted with blue circles, reflecting a stronger recall but lower precision, LLM-Pre = 15.38%, LLM-Rec = 80.00%, and LLM-F1 = 25.80% .

3. DETAILED EXPERIMENT RESULTS

QFESum consists of eight thematic events: **`BPoil`** (BP Oil Spill, from T17), **`finan`** (Global Financial Crisis, from T17), **`iraq`** (Iraq War, from T17), **`syria_t17`** (Syrian Conflict, from T17), **`egypt`** (Egyptian Revolution, from CRISIS), **`libya`** (Libyan Civil War, from CRISIS), **`syria_crisis`** (Syrian Conflict, from CRISIS), and **`yemen`** (Yemeni Uprising, from CRISIS). This section provides detailed per-corpus evaluation results for all experiments.

3.1. **Detailed Gold Documents Summarization Results.** Table 1 reports the number of events of different summarization methods. Table 2 reports ROUGE-1/2/L, BERTScore, and LLM-F1 for the gold document summarization. Here, LLM1-F1 denotes scores evaluated by DeepSeek-v3, while LLM2-F1 denotes scores evaluated by Qwen3-32B.

3.2. **Detailed Retrieved Documents Summarization Results.** Table 3 presents detailed summarization and retrieval results corresponding to the retrieved-document experiments reported in the main paper. Due to space limitations, we report the following evaluation metrics: ROUGE-1/2/L and LLM-F1 (where LLM-F1 is computed using DeepSeek-v3) for summarization performance, and F1 and Hit@5 for retrieval performance.

3.3. **Detailed Ablation Studies Results.** Table 4 presents the detailed results of the ablation study. We report metrics including ROUGE-1/2/L, BERTScore, and LLM-F1 computed using DeepSeek-v3.

4. DETAILED EXPERIMENT SETTINGS

This section describes the detailed configurations used in our experiments.

4.1. **Experimental Setup.** DeepSeek-v3 and ChatGPT were accessed through API calls, while all other experiments were conducted on a MacBook Pro equipped with an Apple M3 Pro chip and 18GB memory. During dataset construction, constructing each query required approximately 2 hours of human effort and 4 hours of program runtime. During summarization, generating the summary for each query took approximately 4 hours. The dataset contains 104 queries in total.

4.2. **Retrieval Arguments Settings.** In the retrieval stage, our RAT model does not require explicit hyperparameter settings. For comparison baselines, we configure the retrieval modules as follows:

BM25 is implemented under a bag-of-words retrieval framework to identify documents semantically relevant to a query in long event-centric collections. Term frequency, document frequency, and inverse document frequency (IDF) are computed over the entire corpus. We adopt the default hyperparameters $k = 1.5$ and $b = 0.75$ for BM25 scoring. For each query, Top-K documents are retrieved based on BM25 scores, where $K = 200$, with a minimum score cutoff of 0.2.

Methods + Input Documents	Corpus	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore	LLM1-F1	LLM2-F1
SHC + $\mathcal{D}_i^{\mathrm{r}}$	BPoil	38.21	10.68	16.1	83.53	10.15	23.76
	finan	32.95	7.45	13.64	82.23	10.21	21.09
	iraq	39.37	10.28	17.88	82.66	11.45	19.34
	syria_t17	40.02	11.73	17.09	84.31	16.85	23.04
	egypt	40.07	8.70	15.42	83.26	16.91	22.71
	libya	38.04	10.37	16.19	83.99	7.09	17.09
	syria_crisis	42.39	14.16	16.42	84.29	13.20	23.59
	yemen	42.47	12.24	17.74	84.24	11.82	31.42
Topic_TLS + $\mathcal{D}_i^{\mathrm{r}}$	BPoil	32.14	8.51	14.03	82.59	6.95	13.46
	finan	33.40	6.66	13.40	82.13	3.27	9.41
	iraq	29.69	5.75	12.29	81.59	1.14	6.28
	syria_t17	28.02	5.08	11.61	81.96	2.56	10.46
	egypt	26.06	5.02	11.56	81.94	2.42	8.81
	libya	26.69	6.37	11.12	82.40	3.02	8.60
	syria_crisis	32.53	6.82	13.01	82.51	5.01	9.26
	yemen	31.29	6.60	13.24	82.58	4.14	11.32
GraphRAG + $\mathcal{D}_i^{\mathrm{r}}$	BPoil	29.93	6.08	15.13	80.41	5.26	17.47
	finan	28.10	4.82	13.64	80.44	7.11	14.01
	iraq	30.25	5.59	14.59	81.61	12.11	14.01
	syria_t17	28.72	5.05	13.83	82.26	20.58	12.78
	egypt	24.54	3.14	12.37	80.75	5.02	7.61
	libya	25.51	4.20	13.43	81.06	3.58	6.71
	syria_crisis	30.60	5.54	14.29	82.77	10.72	19.91
	yemen	24.39	4.35	12.51	81.40	4.68	12.89
FG-RAG + $\mathcal{D}_i^{\mathrm{r}}$	BPoil	31.16	8.75	14.43	82.05	10.72	23.63
	finan	27.31	5.80	12.72	80.84	8.05	16.70
	iraq	31.32	8.15	14.32	82.40	13.76	27.20
	syria_t17	25.89	6.30	11.92	82.50	11.55	20.45
	egypt	24.39	5.76	11.63	81.96	11.99	21.96
	libya	24.60	5.84	11.80	82.49	8.43	18.69
	syria_crisis	28.64	6.53	13.10	83.05	11.48	27.76
	yemen	23.70	5.54	10.97	81.99	9.66	23.33
UnstructBase + $\mathcal{D}_i^{\mathrm{r}}$	BPoil	27.16	6.11	13.32	80.72	3.68	9.90
	finan	22.07	2.99	10.26	79.34	0.50	3.50
	iraq	26.37	7.72	14.75	80.65	4.16	7.99
	syria_t17	25.85	4.71	11.11	81.86	0.00	2.29
	egypt	38.26	5.27	14.29	82.05	0.77	3.49
	libya	23.10	3.67	10.70	80.33	3.41	6.47
	syria_crisis	28.78	4.40	13.34	82.08	0.66	4.67
	yemen	28.69	3.81	12.32	81.03	2.16	6.17

TABLE 2. Per-corpus evaluation of five Methods + Gold Documents on 8 corpus, evaluated by ROUGE-1, ROUGE-2, ROUGE-L, BERTScore, LLM1-F1 and LLM2-F1.

Dense Passage Retrieval (DPR) [10] is implemented using a dual-encoder setup: DPR-question-encoder for queries and DPR-context-encoder for documents (both pretrained on NQ). Documents are split into sliding windows with a maximum length of 512 tokens and stride of 96 for improved coverage. All chunk embeddings are L2-normalized, and the document relevance score is the maximum inner-product (cosine similarity) over all chunks. A threshold-based filtering is applied, where documents with similarity ≥ 0.55 are considered relevant.

RoBERTa-large-MNLI is used as an NLI-based semantic retrieval model. Long documents are encoded with the same sliding-window strategy (max length 512, stride 96). For each chunk, we construct

Methods + Input Documents	Corpus	Summary Metrics				Retrieval Metrics	
		ROUGE-1	ROUGE-2	ROUGE-L	LLM-F1	F1	hit@5
SHC + **RAT**	BPoil	40.67	11.81	17.10	12.60	0.4952	1.0000
	finan	27.56	5.73	11.82	2.29	0.5486	1.0000
	iraq	32.08	6.09	13.77	4.04	0.5564	1.0000
	syria_t17	34.35	8.62	14.85	10.57	0.8152	1.0000
	egypt	39.19	8.38	14.85	6.47	0.5801	1.0000
	libya	34.44	8.88	15.27	8.36	0.4606	1.0000
	syria_crisis	35.23	10.12	14.43	10.92	0.5134	1.0000
	yemen	36.93	9.59	15.89	11.11	0.5325	0.8750
SHC + BM25	BPoil	38.63	10.76	16.54	9.99	0.4789	1.0000
	finan	30.49	6.95	11.94	3.47	0.5229	1.0000
	iraq	33.19	6.14	14.82	2.19	0.5089	1.0000
	syria_t17	27.45	6.26	13.14	5.32	0.6826	1.0000
	egypt	37.20	7.64	15.05	8.95	0.2975	1.0000
	libya	35.28	8.09	15.21	5.33	0.2919	1.0000
	syria_crisis	38.00	8.75	15.40	6.63	0.2126	1.0000
	yemen	36.17	8.57	15.46	4.79	0.2133	0.8750
SHC + DPR	BPoil	41.27	12.07	17.07	9.42	0.3871	0.7619
	finan	23.51	4.72	10.12	2.27	0.4343	0.8095
	iraq	29.96	5.45	12.99	5.15	0.3669	1.0000
	syria_t17	18.93	3.18	10.04	1.54	0.3179	1.0000
	egypt	34.05	6.79	14.77	6.38	0.4080	0.9000
	libya	35.85	8.35	14.84	4.53	0.3334	0.5714
	syria_crisis	37.30	11.18	14.71	8.38	0.3891	1.0000
	yemen	35.32	9.41	14.88	8.76	0.4004	0.6250
SHC + Roberta-mnli	BPoil	13.45	2.36	8.36	2.22	0.0693	0.9048
	finan	14.10	2.74	8.25	0.56	0.0657	0.8571
	iraq	19.48	3.50	10.16	0.00	0.0893	0.9286
	syria_t17	11.38	2.13	6.78	0.00	0.1496	1.0000
	egypt	29.20	6.51	13.13	5.46	0.0963	0.9000
	libya	39.86	11.19	15.90	13.43	0.0811	0.4286
	syria_crisis	31.73	6.33	13.87	2.42	0.0674	0.7500
	yemen	23.63	4.47	10.78	2.90	0.0507	0.6250

TABLE 3. Per-corpus evaluation across retrieval baselines using Summary Metrics (ROUGE-1/2/L, LLM-F1) and Retrieval Metrics (F1, hit@5).

three "relevant" templates and one "irrelevant" template by pairing the document with the query in natural language form. Predictions are classified into entailment (ENT), neutral (NEU), or contradiction (CONTRA). We adopt a *semantic_strict* rule: a document is labeled as relevant if *any* chunk entails *any* relevant template (predicted as ENT).

These configurations ensure a fair and consistent evaluation of retrieval effectiveness across all models.

4.3. **Summarization Arguments Settings. Event Extraction Settings:** As shown in Figure 2, we present the Prompt Used for Event Extraction in our framework.

SHC Arguments Settings:

Methods	Corpus	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore	LLM-F1
SHC + $\mathcal{D}_i^r$	BPoil	38.21	10.68	16.1	83.53	10.15
	finan	32.95	7.45	13.64	82.23	10.21
	iraq	39.37	10.28	17.88	82.66	11.45
	syria_t17	40.02	11.73	17.09	84.31	16.85
	egypt	40.07	8.70	15.42	83.26	16.91
	libya	38.04	10.37	16.19	83.99	7.09
	syria_crisis	42.39	14.16	16.42	84.29	13.20
	yemen	42.47	12.24	17.74	84.24	11.82
SHC + RAT	BPoil	40.67	11.81	17.10	84.00	12.60
	finan	27.56	5.73	11.82	81.76	2.29
	iraq	32.08	6.09	13.77	82.28	4.04
	syria_t17	34.35	8.62	14.85	83.33	10.57
	egypt	39.19	8.38	14.85	83.21	6.47
	libya	34.44	8.88	15.27	83.62	8.36
	syria_crisis	35.23	10.12	14.43	84.27	10.92
	yemen	36.93	9.59	15.89	83.50	11.11
w/o hierarchies	BPoil	34.21	9.19	15.59	83.32	8.75
	finan	29.13	6.54	12.01	82.43	3.48
	iraq	26.10	4.80	12.59	82.13	2.30
	syria_t17	37.20	9.42	15.94	84.41	7.61
	egypt	36.23	6.81	14.47	83.53	8.22
	libya	39.12	9.95	16.53	84.20	6.14
	syria_crisis	36.09	10.25	14.59	84.37	8.05
	yemen	41.34	11.93	17.07	84.93	13.50
w/o retrieval	BPoil	38.81	11.32	16.15	83.53	9.61
	finan	29.50	7.03	11.56	82.26	2.54
	iraq	34.07	7.04	14.63	82.46	3.74
	syria_t17	33.09	9.08	14.22	83.85	6.41
	egypt	36.59	7.98	13.90	83.33	11.10
	libya	29.30	7.89	12.78	83.25	6.08
	syria_crisis	29.81	9.26	12.42	83.82	10.05
	yemen	34.85	9.78	13.91	83.35	6.31
w/o retrieval&hierarchies	BPoil	36.13	10.11	15.69	83.20	9.55
	finan	27.53	6.63	10.81	81.85	1.81
	iraq	30.20	5.18	13.23	82.11	1.39
	syria_t17	32.82	7.18	13.48	83.63	5.94
	egypt	35.28	7.47	13.72	83.20	6.02
	libya	31.94	8.62	13.78	83.49	5.14
	syria_crisis	26.58	8.73	11.16	83.76	8.11
	yemen	33.55	10.42	14.19	83.96	8.94

TABLE 4. Ablation study across eight corpus comparing the full model and ablated variants (w/o hierarchies, w/o retrieval, w/o retrieval&hierarchies), evaluated on ROUGE-1/2/L, BERTScore and LLM-F1.

Due to substantial variations across corpus in terms of the number of original documents (*gold_documents_num*), extracted events (*gold_documents_event_num*), and reference summary event number (*reference_summary_event_num*)—as presented in Table 6—the difficulty of the summarization task can differ significantly with respect

Prompt for Event Extraction

Role: Event Extraction Assistant

General Description. You are an Event Extraction assistant. Your task is to extract all events that are strictly relevant to a given query {query} from the text. The input document is a news article or report, and the query represents a specific topic of interest. Do not explain your reasoning, only output the final JSON.

Instructions:

- Read the provided text.
- Extract all **events** that are directly related to the query.
- Each event should be a standalone sentence or phrase that captures what happened, who did it, and any consequence if available.
- Only include events that are clearly connected to the query.
- Event retrieval mainly focuses on the sentence level. You may refine incomplete or overly long sentences using context.
- Output the results in the JSON format specified below.

Output Format:

```
{
  "events": [
    "First relevant event.",
    "Second relevant event.",
    ...
  ]
}
```

Final Instruction. Now, use this format to process the following document:

Query: {query}

Text: {text}

Output:

FIGURE 2. Prompt for Event Extraction

to compression ratio and information density. To mitigate the influence of such input-scale imbalance on output length, we introduce three controllable parameters during the BERTopic clustering in SHC: *min_dist*, *min_cluster_size*, and *min_samples*, whose configurations are detailed in Table 5. Specifically, *min_dist*, *min_cluster_size*, and *min_samples* influence semantic compactness and valid cluster formation in the second-stage clustering.

It is worth emphasizing that parameter tuning in this study is guided by a practical objective—ensuring that the generated summaries remain within a reasonable length range—rather than corpus-specific hyper-optimization. With this lightweight global adjustment, the average generated summary event number across the eight corpus are 22.79, 41.62, 11.14, 20.62, 26.00, 24.71, 38.12, and 25.00 respectively. Although some variations remain, the outputs are overall maintained at a comparable scale, preventing compression-ratio disparity.

Summarization Baselines Settings:

Topic_TLS: Clustering events via LLM-based similarity and selects key representatives. We adopt Qwen2.5-7B to extract query-focused events, following the clustering configuration in [9]: for T17, the

Prompt for Citation Evaluation

Role: Data Annotator
Instructions:
You are provided with the following materials:
- **Passage**: passage
- **Sentence**: sentence
Task: Assess whether the passage fully supports the sentence.
Choices:
1. **Fully Supports**: Select this option if the passage completely and clearly supports every aspect of the sentence.
2. **Does Not Fully Support**: Select this option if any discrepancies, omissions, or inaccuracies in the passage prevent it from fully supporting the sentence.
Output:
- If the passage fully supports the sentence, output "Yes."
- If it does not, output "No."
Note: Please refrain from adding any content not requested in the instructions.

FIGURE 3. Prompt for LLM-based evaluation

top 36 clusters are selected and 3 events sampled from each; for CRISIS, the top 29 clusters are selected and 2 events sampled from each.
GraphRAG: Input original documents rather than event list, building a knowledge graph for hierarchical community summarization. Implemented using `ChatGPT-4o`-mini with all default parameters.
FG-RAG: Augments GraphRAG with context-aware entity expansion to improve retrieved coverage. Implemented using Qwen2.5-7B with all default parameters.
UnstructBase: While proposing this evidence-extraction-based summarization paradigm, the original paper also introduced the SUnSet dataset to improve summarization quality. However, due to resource limitations, this work only follows the paradigm during reproduction and does not conduct additional training; instead, we use the Qwen2.5-7B model directly. Since our method also involves no training procedure, this reproduced paradigm still provides a meaningful comparison with our approach. All other parameters are set to the default values reported in the original paper.

4.4. **LLM Evaluation Settings.** Specifically, both the generated summary and the reference summary are first segmented into sentences, with each sentence treated as an individual event. For each generated sentence, we iterate over all reference sentences and employ an LLM to determine whether a semantic alignment exists. Precision measures the correctness of the generated summary, Recall reflects their coverage of reference summary, and F1 provides a balanced overall assessment. For the judging models, we adopt DeepSeek-v3 [14] (671B parameters, strong semantic capability) as the primary evaluator, and use Qwen3-32B [26] as an auxiliary evaluator to enhance robustness and reliability.

The prompt used in this work follows the template proposed in [21], as shown in Figure 3. In the original study, this prompt was designed to determine whether one sentence semantically supports another. Here, we extend its use to evaluate whether two sentences refer to the same event, which aligns well with the event-level semantic matching requirement in QFES. Therefore, we adopt this template directly for LLM-based evaluation.

5. DETAILED QFESUM SETTINGS

The Figure 4 shows the *query generation* prompt P_{pg} and an example. The Figure 5 shows the *instruction generation* prompt P_{ig}. The Figure 6 shows the *relevance judgement* prompt P_{rj}. The

Prompt for Query Generation
Instructions:
You are given a set of topic keywords extracted from a semantic cluster. Your task is to integrate these keywords into a concise, fluent, and human-readable query phrase that summarizes the underlying semantic theme. The output should be suitable for use as a query in a query-focused summarization task.
Input Keywords: {*"revolution," "revolutions," "youth," "revolutionaries," "regimes," "dictators"*}
Output Query Phrase: {*"Youth-led revolutionary movements"*}

FIGURE 4. Prompt used Query Generation

Corpus	min_dist	min_cluster_size	min_samples
BPoil	0.05	14	13
finan	0.005	3	2
iraq	0.05	10	4
syria_t17	0.05	15	15
egypt	0.05	15	15
libya	0.05	19	19
syria_crisis	0.05	19	23
yemen	0.05	19	19

TABLE 5. Parameter settings for each corpus, including min_dist, min_cluster_size, and min_samples.

Figure 7 shows an example of the query "Treasury Department response and intervention" relevance judgement prompt P_{rj} that ChatGPT-4o generated to guide Qwen2.5-7B to annotate sentence-level relevance.

Prompt for Instruction Generation
Instructions:
You are given a query and a reference summary focused on this query. All sentences in the summary are manually selected as relevant to the query. Your task is to carefully read the reference summary and, in conjunction with the query, produce a prompt that helps a large language model judge whether a given sentence is relevant to the query. The prompt should include (i) a detailed semantic clarification of the query and (ii) illustrative positive and negative example sentences.
Input Query: {*Query*}
Input QFS Summary: {*QFS Summary*}
Output Annotation Prompt: {*Annotation Prompt*}

FIGURE 5. Prompt for Instruction Generation

Prompt for Relevance Judgment
Instructions:
Please determine whether the following sentence is strictly relevant to the query: "query". Only answer "True" if the sentence explicitly discusses issues related to the query. Otherwise, respond "False".
Sentence: "sent"
Answer with a single word: "True" or "False".
Input Query: {*Query*}
Input Sentence: {*Sentence*}
Output Labels: {*True or False*}

FIGURE 6. Prompt for Relevance Judgment

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Index	BPoil	finan	iraq	syria_t17	egypt	libya	syria_crisis	yemen
gold_documents_num	222.57	75.19	82.07	286.12	786.70	599.29	1146.22	899.38
gold_documents_event_num	875.81	275.48	277.50	1280.25	2176.60	1365.14	2959.11	1884.38
reference_summary_event_num	21.43	28.43	16.00	30.12	18.40	17.36	21.62	18.88
generated_summary_event_num	22.79	41.62	11.14	20.62	26.00	24.71	38.12	25.00

TABLE 6. Documents and summary event number across the eight corpus.

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Prompt for Relevance Annotation

You are given a query and a sentence. Your task is to classify whether the sentence is relevant to the query. The following are some guidance about the task.

Query: "Treasury Department response and intervention"

This query refers to actions, policies, announcements, or decisions made by the query "`U.S. Department of the Treasury`" in response to the 2008 financial crisis. Relevant sentences include:

- Statements or public addresses by **Treasury Secretary Henry Paulson**
- Treasury-led programs like the **$700 billion financial rescue (TARP)** and other bailout efforts
- Decisions to inject capital into banks, buy mortgage-backed securities, or stabilize money-market funds
- Appointments or hiring of officials (e.g., Neel Kashkari) or legal/financial advisors to oversee intervention programs
- Testimony or defenses of Treasury actions before Congress
- Joint actions with the Federal Reserve or FDIC where Treasury plays a lead role
- Any Treasury announcements regarding guarantees, investments, or expanded rescue efforts

Here are examples of relevant sentences:

(1) "Henry Paulson says the government will start buying securities backed by mortgages – initially, up to $5 billion worth."
(2) "Treasury Secretary Henry Paulson urges patience as global markets continue to slide."
(3) "Treasury Secretary Paulson lays out steps to shore up the nation's banking system by having the government buy shares in leading banks."
(4) "Treasury Dept. says it will announce that a group of large regional banks have agreed to accept investments from the government, joining nine of the largest American banks who were forced to accept $125 billion in funding last week."
(5) "The Treasury and FDIC say they are crafting a plan under which the government would guarantee the mortgages of as many as 3 million homeowners now struggling to avoid foreclosure."

Here are examples of sentences that are irrelevant:

(1) "The government seized control of the insurance giant American International Group to preserve a crucial bulwark of the global financial system."
(2) "The Federal Reserve was forced to ask the Treasury yesterday to borrow some extra money to replenish its coffers."
(3) "Democrats want to create a new agency that would use money borrowed by the Treasury to recapitalize troubled financial institutions by buying some of their unwanted loans and securities at discounted prices."
(4) "House Speaker Nancy Pelosi dispatched Rep. Barney Frank to determine whether Federal Reserve Chairman Ben S. Bernanke should retain authority to unilaterally bail out failing firms."
(5) "Bush has left direct management of the crisis to them and other advisers, and has limited his public remarks on the economy."

Sentences are **not relevant** if they only describe the Federal Reserve or the government's response to the economic crisis, or if they only describe the requests and demands of other institutions or departments to the Treasury Department, without describing the measures and actions taken by the Treasury Department.

Sentences **are relevant** only if they directly refer to the measures and actions taken by the Treasury Department.

Now determine whether the following sentence is strictly relevant to the query.

Query: "`{query}`"

Sentence: "`{sentence}`"

Respond with only:

`true` or `false`

Do not include any explanation or additional text.

FIGURE 7. Prompt for Relevance Annotation

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